
Magnetic flux quantisation and vacuum counterterms in real-time lattice simulations of relativistic-quantum plasmas

Anonymous

Abstract

Real-time lattice quantum electrodynamics promises a first-principles description of plasmas in which collective and Compton scales overlap, but the regime in which its classical-statistical formulation is actually predictive has not been mapped. Working with a variational Yee-type discretisation of scalar QED whose discrete $U(1)$ symmetry conserves charge exactly, we show that the framework is quantitatively faithful in two previously untested strong-coupling regimes and structurally obstructed in two others. In a plasma dressed by a relativistically strong circularly polarised wave the lattice reproduces, up to $a = 8$, the Volkov mass $m_* = m\sqrt{1 + a^2}$ and the dressed plasma frequency $\omega_p(1 + a^2)^{-1/4}$ to 0.98%, and with them the opposite shifts of the electromagnetic cutoff and of the pair-creation gap $2m_*$, the latter to 0.73%; for counter-streaming relativistic boson beams it reproduces the growth rate of the quantum two-stream instability to 1.08%, including three unstable wavenumbers that lie entirely outside the classical cold-fluid unstable band, while conserving each species' charge to 1.6×10^{-15} . Two obstructions are then identified and removed. First, no single-valued periodic gauge potential can carry net magnetic flux on the torus, so a uniform magnetic field — and with it the entire relativistic Landau ladder — is unrepresentable: we measure $\langle B_z \rangle = 0$ to machine precision and level errors of 5.1% to 10.9% that do not decrease with resolution. Restoring 't Hooft twisted boundary conditions quantises the flux, $qB_0L_xL_y = 2\pi N$, recovers the ladder to 0.18% at every resolution, and preserves exact charge conservation and the Bianchi identities. Second, seeding every lattice mode with half a quantum — the prescription required to describe pair creation out of the vacuum — gives the charged boson a Rayleigh–Jeans self-energy that diverges logarithmically with the cutoff, so the physical mass drifts by 10.62% between resolutions and the theory has no continuum limit; a band-limited fluctuation sector plus a gauge-invariant lattice mass counterterm removes the drift to 0.15%. The two obstructions share a structure: the scheme is exact where the field configuration is topologically trivial and the occupation numbers are large, and fails, curably, where either condition is violated.

1 Introduction

The plasma-kinetic description of matter is a geometrical-optics limit of an underlying relativistic-quantum field theory [2]. It fails when the collective scales of a plasma stop being large compared with the Compton wavelength of its constituents: in electron–positron pair plasmas created by petawatt lasers [9, 8], in the pair cascades expected at multi-petawatt facilities and beam–laser collision experiments [3, 4, 5, 6], and in the magnetospheres of magnetars, where the field strength approaches the Schwinger value $E_c = m^2/q$ at which the vacuum itself becomes unstable to pair creation [13, 14, 15]. The elementary process behind those cascades, light-by-light pair creation [12], was first observed at SLAC E144 [11] and is about to be studied in the strongly nonlinear regime at dedicated

beam–laser facilities [7]. Semiclassical treatments patch this by inserting prefabricated source terms into kinetic or fluid equations [18, 19, 20], but such terms cannot describe the interference between coexisting channels [21], and they provide no systematic way to remove energy and momentum from the field once many pairs have been produced.

A first-principles alternative exists. When occupation numbers are large and the coupling is weak — the situation in essentially every laboratory plasma, where $q \simeq 0.3$ — the quantum evolution of a gauge theory is dominated by tree-level processes and is reproduced by advancing the *classical* field equations over an ensemble of statistically equivalent initial conditions [25, 26, 27, 28, 29, 30, 31, 32, 33, 39]. The same construction is the truncated Wigner approximation of ultracold-atom physics [36, 37, 38], and its ultraviolet behaviour — the subject of one of our results — has been analysed at the level of the loop expansion by Epelbaum, Gelis and collaborators [34, 35]. Applied to lattice gauge theory this “classical-statistical” approach has described string breaking and fermion production in 1+1-dimensional QED [40, 41, 42], and Shi, Xiao, Qin and Fisch [1] extended it to plasma physics by discretising the scalar-QED action on a Yee lattice in a way that respects the geometric structure of discrete exterior calculus. The construction is attractive because its invariants are exact rather than approximate: the discrete $U(1)$ gauge symmetry of the action makes local charge conservation a machine-precision identity, and $\nabla \cdot \mathbf{B} = 0$ and Faraday’s law hold by construction because the field strength is the discrete exterior derivative of the gauge connection. Related structure-preserving discretisations of the Maxwell–Klein–Gordon system [84, 85] and geometric particle-in-cell schemes [86, 88, 89] share this philosophy.

The real-time lattice programme has developed quickly since. Improved and implicit real-time actions have been constructed [44, 48, 47]; canonical symplectic integrators for lattice strong-field QED have been derived [46] in the same structure-preserving spirit as geometric particle-in-cell [87, 91, 90]; tensor-network [45] and cold-atom [43, 92, 93] analogue simulators have been demonstrated; and quantum-computing routes to the same ensembles are being explored [57, 58]. On the plasma side the strong-field regime has been mapped with kinetic and Wigner-function methods [54, 50, 51, 52, 53, 56] and reviewed in [49].

What has not been established is the boundary of the regime in which the scheme is predictive. The demonstrations in [1] — linear waves in an unmagnetised single-species plasma, and a gamma-ray laser incident on a dense slab — both live in a corner of the framework’s claimed scope: the gauge field is topologically trivial, the matter field is a highly occupied condensate, and the only strong-field physics is a longitudinal wakefield. Magnetised plasmas, relativistically dressed plasmas, kinetic beam instabilities and genuine vacuum physics are all advertised applications and none of them was tested. This matters because the two structural ingredients that make the scheme attractive — the compactness of the gauge connection and the classical sampling of quantum fluctuations — are precisely the ingredients that can fail, and they fail in ways that produce plausible-looking output rather than obvious numerical breakdown.

In this paper we map that boundary and repair it. We first re-derive and re-implement the variational discretisation from the action, verifying that charge conservation, the Bianchi identities and gauge invariance hold to machine precision (Sec. 2), and we reproduce both published demonstrations as an implementation control (Sec. 4.1). We then report four results.

Two of them extend the verified reach of the framework into regimes where the matter–gauge coupling is strongly nonlinear. In a plasma illuminated by a homogeneous, relativistically strong, circularly polarised wave the charged bosons acquire the Volkov mass $m_* = m\sqrt{1 + a^2}$, with $a = qA_0/m$; we show that the lattice reproduces not only m_* but the entire reorganisation of the wave spectrum that follows from it — the electromagnetic cutoff falls as $\omega_p(1 + a^2)^{-1/4}$ while the pair-creation gap rises as $2m\sqrt{1 + a^2}$ — over two decades in the intensity parameter. In a pair of counter-streaming relativistic boson beams we derive the multi-beam cold electrostatic dispersion relation of scalar QED, validate it by Lorentz covariance and against the single-species limit, and show that the lattice reproduces the resulting growth rates, including at wavenumbers where the classical relativistic cold-fluid theory predicts complete stability. Quantum recoil therefore widens the unstable band of the two-stream instability by more than 75% in this regime, and the lattice resolves the difference.

The other two results are obstructions. The first is topological. On a periodic domain, the plaquette sum of $B_z = \partial_x A^y - \partial_y A^x$ telescopes to zero for *any* single-valued gauge potential, so a uniform magnetic field simply does not exist in the scheme’s configuration space. We show that the resulting failure is not a discretisation artefact — it is independent of Δx , and every naive workaround either

concentrates the flux into a one-cell seam that radiates or replaces the uniform field by one with zero mean — and that the cure is to abandon strict periodicity in favour of ’t Hooft twisted boundary conditions [59, 60, 61], in which the fields are periodic up to a transition function. The price is that the field becomes quantised, $qB_0L_xL_y = 2\pi N$; the gain is that the relativistic Landau ladder $E_n^2 = m^2 + (2n+1)qB_0$ is recovered to better than 0.18% while charge conservation, the Bianchi identities and gauge invariance remain exact.

The second obstruction is ultraviolet. To describe pair creation from the vacuum, the classical-statistical prescription requires that every lattice mode be seeded with half a quantum. That seeding deposits a classical energy density that grows like Δx^{-2} , and it gives the charged boson a Rayleigh–Jeans self-energy $\delta m^2 = q^2 \langle A_\perp^2 \rangle_{\Delta x}$ that diverges logarithmically as the lattice is refined. The physical mass, and with it the plasma frequency, the light cone and the Schwinger threshold, therefore depend on the lattice spacing: the theory as formulated has no continuum limit. This is the lattice manifestation of the known non-renormalisability of the classical-statistical approximation [34], and normal ordering — the quantum-field-theoretic cure — is unavailable because the “vacuum” here is an actual sampled field configuration. We show that band-limiting the sampled fluctuations and adding a gauge-invariant lattice mass counterterm to the discrete Lagrangian restores a continuum limit, and we state precisely what the modification does and does not fix.

Taken together the four results identify a single organising principle. The variational lattice scheme is exact — in the strong sense that its invariants are machine-precision identities and its predictions match analytic theory at the sub-percent level — whenever the gauge configuration is topologically trivial and the matter occupation numbers are large. Both obstructions we find are violations of exactly one of those two conditions, and both are removed by modifications that operate at the level of the discrete action rather than at the level of the solver.

2 Methodology

2.1 Discrete action and equations of motion

We work in natural units $\hbar = c = \epsilon_0 = 1$ and measure lengths in $1/m$ and frequencies in m , where m is the boson mass. The continuum theory is scalar QED,

$$\mathcal{L} = \overline{(D_\mu \phi)}(D^\mu \phi) - m^2 \bar{\phi}\phi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \quad D_\mu = \partial_\mu - iqA_\mu, \quad (1)$$

whose Euler–Lagrange equations are the Klein–Gordon–Maxwell (KGM) system. Following [1] we discretise the *action*, not the equations. The complex scalar ϕ_v lives on vertices, the gauge one-form A_e on edges, and the field strength $F_f = dA$ on faces. In 1+1 dimensions with the two transverse polarisations retained as a dimensional reduction (all fields independent of y, z ; A^y, A^z then sit on vertices), the discrete action is

$$S_d = \Delta x \Delta t \sum_{n,i} \left\{ |D_0 \phi|^2 - |D_x \phi|^2 - (m^2 + q^2(A^{y2} + A^{z2}))|\phi|^2 + \frac{1}{2}(E_x^2 + E_y^2 + E_z^2 - B_y^2 - B_z^2) \right\}, \quad (2)$$

with pull-back covariant differences built from Wilson lines,

$$(D_0 \phi)_i^{n+1/2} = \frac{\phi_i^{n+1} V_i^{n+1/2} - \phi_i^n}{\Delta t}, \quad (D_x \phi)_{i+1/2}^n = \frac{L_{i+1/2}^n \phi_{i+1}^n - \phi_i^n}{\Delta x}, \quad (3)$$

$V_i^{n+1/2} = e^{-iq\Delta t \Phi_i^{n+1/2}}$, $L_{i+1/2}^n = e^{+iq\Delta x A_{i+1/2}^{x,n}}$, and with $E_{i+1/2}^{x,n+1/2} = -(A^{x,n+1} - A^{x,n})_{i+1/2}/\Delta t - (\Phi_{i+1} - \Phi_i)^{n+1/2}/\Delta x$, $E_i^{y,n+1/2} = -(A_i^{y,n+1} - A_i^{y,n})/\Delta t$, $B_{i+1/2}^{z,n} = (A_{i+1}^y - A_i^y)^n/\Delta x$ and $B_{i+1/2}^y = -(A_{i+1}^z - A_i^z)^n/\Delta x$. Note that Eq. (2) uses the Yee form $E^2 - B^2$ [24] of the Maxwell term rather than Wilsonian plaquettes [23], which is what makes the discrete local energy conservation law exact in the $q \rightarrow 0$ limit [1].

Every equation we integrate is a variation of Eq. (2). Setting $\delta S_d / \delta \bar{\phi}_i^n = 0$ gives the discrete Klein–Gordon equation, whose explicit form in the temporal gauge $\Phi = 0$ used throughout is

$$\phi_i^{n+1} = 2\phi_i^n - \phi_i^{n-1} + C^2 (L_{i+1/2}^n \phi_{i+1}^n - 2\phi_i^n + \bar{L}_{i-1/2}^n \phi_{i-1}^n) - \Delta t^2 (m^2 + \delta m^2 + q^2 A_\perp^2) \phi_i^n, \quad (4)$$

with $C = \Delta t / \Delta x$ and δm^2 a counterterm that is zero except where stated in Sec. 2.4. Setting $\delta S_d / \delta \Phi_i^{n+1/2} = 0$ gives the discrete Gauss law $\nabla \cdot \mathbf{E} = \rho + \rho_b$ with

$$\rho_i^{n+1/2} = \frac{2q}{\Delta t} \text{Im}[\bar{\phi}_i^n \phi_i^{n+1} V_i^{n+1/2}], \quad (5)$$

and ρ_b a static neutralising ion background. Setting $\delta S_d / \delta A_{i+1/2}^{x,n} = 0$ and $\delta S_d / \delta A_i^{y,n} = 0$ gives the Maxwell–Ampère laws

$$\frac{E^{x,n+1/2} - E^{x,n-1/2}}{\Delta t} = j^{xn}, \quad \frac{E^{y,n+1/2} - E^{y,n-1/2}}{\Delta t} = -(\partial_x B^z)^n + 2q^2 A^{y,n} |\phi^n|^2, \quad (6)$$

with the gauge-invariant lattice current

$$j_{i+1/2}^{x,n} = \frac{2q}{\Delta x} \text{Im}[\bar{\phi}_i^n L_{i+1/2}^n \phi_{i+1}^n]. \quad (7)$$

Because Eq. (5) and Eq. (7) are the variational derivatives of the *same* action that gives Eq. (4), Noether’s theorem applied to the discrete $U(1)$ symmetry $\phi_i^n \rightarrow \phi_i^n e^{iq\alpha_i^n}$, $A_e \rightarrow A_e + d\alpha$ yields the exact lattice continuity equation $(\rho^{n+1/2} - \rho^{n-1/2}) / \Delta t = (j_{i+1/2}^x - j_{i-1/2}^x) / \Delta x$, so the Gauss law is preserved and the total charge is conserved to round-off. The time advance is fully explicit: $E^{n+1/2}$ from Eq. (6), ϕ^{n+1} from Eq. (4), then $A^{n+1} = A^n - \Delta t E^{n+1/2}$.

2.2 Two spatial dimensions and twisted boundary conditions

Magnetic fields require two spatial dimensions. The same construction gives

$$S_d = \Delta x \Delta y \Delta t \sum_{n,i,j} \left\{ |D_0 \phi|^2 - |D_x \phi|^2 - |D_y \phi|^2 - m^2 |\phi|^2 + \frac{1}{2} (E_x^2 + E_y^2 - B_z^2) \right\}, \quad (8)$$

with $B_{i+1/2,j+1/2}^z = (A_{i+1,j+1/2}^y - A_{i,j+1/2}^y) / \Delta x - (A_{i+1/2,j+1}^x - A_{i+1/2,j}^x) / \Delta y$ on plaquettes. Varying gives $\partial_t E^x = \partial_y B^z + j^x$, $\partial_t E^y = -\partial_x B^z + j^y$ and the two-dimensional Klein–Gordon equation, and B^z remains an exact discrete curl so that $\partial_t B^z = -(\nabla \times \mathbf{E})_z$ holds identically.

The scheme of [1] imposes strict periodicity. We will need the generalisation in which the fields are periodic only up to a gauge transformation — a transition function, in the sense of ’t Hooft [59]. Crossing the x -boundary applies a gauge parameter $\alpha_x(y)$, crossing the y -boundary applies $\alpha_y(x)$:

$$\begin{aligned} \phi_{i+N_x,j} &= \phi_{i,j} e^{iq\alpha_x(y_j)}, & A_{i+N_x,j+1/2}^y &= A_{i,j+1/2}^y - \partial_y \alpha_x, \\ \phi_{i,j+N_y} &= \phi_{i,j} e^{iq\alpha_y(x_i)}, & A_{i+1/2,j+N_y}^x &= A_{i+1/2,j}^x - \partial_x \alpha_y. \end{aligned} \quad (9)$$

Ordinary periodicity is $\alpha \equiv 0$. The transition functions are themselves discrete gauge transformations, so the discrete action (8) is unchanged in form; consequently $U(1)$ gauge invariance, exact charge conservation and the Bianchi identities all survive verbatim. What does not survive is the freedom to choose the flux: single-valuedness of the transition functions around the torus (the cocycle condition at the corner) requires

$$q B_0 L_x L_y = 2\pi N, \quad N \in \mathbb{Z}. \quad (10)$$

For a uniform field in the Landau gauge $\alpha_x(y) = -B_0 L_x y$, $\alpha_y = 0$; in the symmetric gauge $\alpha_x(y) = -\frac{1}{2} B_0 L_x y$, $\alpha_y(x) = +\frac{1}{2} B_0 L_y x$.

2.3 Invariants used as diagnostics

Four classes of invariant are monitored. (i) *Charge*: the lattice continuity equation and the total charge $Q = \sum_i (\rho_i + \rho_{b,i}) \Delta x$. (ii) *Energy*: the discrete energy density of [1], $\mathcal{H} = \mathcal{H}[\phi] + \mathcal{H}[A]$ with $\mathcal{H}^{n+1/2}[A] = \frac{1}{2} (E^{n+1/2})^2 + \frac{1}{2} B^{n+1} B^n$, whose total $\mathcal{U} = \sum_i \mathcal{H}_i \Delta x$ is conserved up to $\mathcal{O}(qn\Delta t^2)$. (iii) *Bianchi*: the discrete Faraday law. (iv) *Gauge invariance*: identical E , B , ρ , j under a random discrete gauge transformation. Table 1 of Sec. 3 reports all four for the 1D and 2D implementations.

2.4 Analytic predictions

All predictions below are stated before any simulation result is quoted, and all are evaluated from formulas at analysis time; no reference value is hard-coded in the simulation loop.

Lattice symbols. We write $\hat{\omega} = 2 \sin(\omega \Delta t / 2) / \Delta t$ and $\hat{k} = 2 \sin(k \Delta x / 2) / \Delta x$. Inserting a plane wave into the free discrete Klein–Gordon equation gives $\hat{\omega}^2 = \hat{k}^2 + m^2$ exactly, and we use the same substitution to turn each continuum prediction into its lattice counterpart. At $m \Delta x = 0.04$ and $k = 4m$ the two differ by 1.1×10^{-3} in $\hat{k}/k$ and by 2.7×10^{-4} in $\hat{\omega}/\omega$ at $m \Delta t = 0.02$; every comparison in this paper is made against the lattice form, so this is never the limiting error, and the convergence studies confirm it.

(P1) Cold linear waves (implementation control). For a single mobile species on an immobile background the transverse electromagnetic branch obeys $\hat{\omega}^2 = \hat{k}^2 + \omega_p^2$ with $\omega_p^2 = q^2 n_0 / m$; this is *exact* on the lattice, because a homogeneous cold condensate has constant modulus under Eq. (4) and the transverse current is $-2q^2 A_\perp |\phi_0|^2$. The electrostatic branches solve $1 + \chi_p = 0$ with the cold scalar-QED susceptibility of [78, 1]. Linearising Eqs. (4)–(7) about $\phi_0 e^{-imt}$ (App. A) we obtain the equivalent polynomial form

$$u^2 - (4m^2 + \omega_p^2)u + 4m^2(\omega_p^2 - k^2) = 0, \quad u = \omega^2 - k^2, \quad (11)$$

whose two positive roots are the Langmuir branch ($\omega \rightarrow \omega_p$ as $k \rightarrow 0$) and the pair branch ($\omega \rightarrow 2m$). We verified numerically that the roots of Eq. (11) coincide with those of $1 + \chi_p = 0$ to 10^{-15} . The poles of χ_p , at $\omega = \sqrt{m^2 + k^2} \mp m$, are the single-particle and pair-continuum resonances; they carry spectral weight but are not roots.

(P2) Wakefield amplitude and the Schwinger threshold (implementation control). For a laser of normalised amplitude $a = qE_L / m\omega_0$ driving a plasma of frequency ω_p , the wave-breaking longitudinal field is $E_x \simeq am\omega_p/q$, i.e.

$$\max |E_x| / E_c = a \omega_p / m. \quad (12)$$

Pair production is expected only when this exceeds the Schwinger field, and should be visible as the appearance of positive charge density (positrons) in a plasma whose charge density is initially negative everywhere.

(P3) Relativistically dressed plasma. Let a homogeneous plasma of charge density qn carry a circularly polarised $k = 0$ pump $A^y + iA^z = A_0 e^{i\omega t}$ with $a = qA_0/m$. Because $A_\perp^2 = A_0^2$ is constant — the property that defines the Volkov states and the intensity-dependent mass [65, 66, 67] — $\phi = \phi_0 e^{-im_* t}$ with

$$m_* = m \sqrt{1 + a^2} \quad (13)$$

is an exact solution of Eq. (4), and the pump obeys $\omega^2 = 2q^2 |\phi_0|^2$. Since $\rho = -2qm_* |\phi_0|^2$, at fixed density $|\phi_0|^2 = n/2m_*$ and the pump frequency is the *dressed* plasma frequency familiar from relativistic self-induced transparency [68, 69, 70, 71]

$$\tilde{\omega}_p = \omega_p (1 + a^2)^{-1/4}, \quad \omega_p^2 = q^2 n / m. \quad (14)$$

Linearising about this state (App. B) shows that probe perturbations at $k \neq 0$ obey the undressed relations with the substitutions $m \rightarrow m_*$ and $\omega_p \rightarrow \tilde{\omega}_p$. We therefore predict, for every probe wavenumber the box resolves up to the spectral limit, $2\pi/L \leq k \leq 3m$,

$$\begin{aligned} \text{(D1)} \quad m_* &= m \sqrt{1 + a^2}; & \text{(D2)} \quad \hat{\omega}_T^2 &= \hat{k}^2 + \tilde{\omega}_p^2; \\ \text{(D3)} \quad \omega_P(k) &\text{ from Eq. (11) with } (m, \omega_p) \rightarrow (m_*, \tilde{\omega}_p), \end{aligned} \quad (15)$$

the last having the $k \rightarrow 0$ gap $2m_*$. We do *not* predict the Langmuir branch: the pump couples transverse perturbations to density perturbations through $2q^2 (\mathbf{A}_\perp \cdot \delta \mathbf{A}_\perp) |\phi|^2$, which mixes the low-frequency longitudinal response with sidebands at $\omega \pm \tilde{\omega}_p$ and is not captured by the substitution above; the observed behaviour is discussed in Sec. 5. One caveat is stated here rather than after the fact. The homogeneous circularly polarised state is an exact solution but not necessarily a stable one: it is subject to a modulational instability whose unstable band lies at small k . The box must therefore be small enough that the unstable band is not resolved, and we verify stability directly by monitoring the drift of the pump amplitude over each run. Section 5 reports what happens when the band *is* resolved. The *primary diagnostic* of this test is the non-trivial ϕ – A coupling: m_* and the pair gap are set entirely by the covariant-derivative terms of Eq. (2), and a scheme that got the coupling right only to leading order in qA would fail at $a \gtrsim 1$.

(P4) Counter-streaming relativistic beams. Linearising Eqs. (4)–(7) about a set of drifting cold condensates $\phi_z = \phi_{0z} e^{i(p_z x - E_z t)}$, $E_z = \sqrt{m_z^2 + p_z^2}$ (App. A) gives the multi-beam electrostatic dispersion relation

$$\omega^2 = \sum_z \tilde{\omega}_{pz}^2 \left[1 + \frac{(k + 2p_z)^2}{2D_{z+}} + \frac{(k - 2p_z)^2}{2D_{z-}} \right], \quad D_{z\pm} = \omega^2 \pm 2E_z \omega - k^2 \mp 2p_z k, \quad (16)$$

with $\tilde{\omega}_{pz}^2 = 2q_z^2 |\phi_{0z}|^2$. This is the scalar-QED counterpart of the multistream models of quantum plasma physics [73, 77, 74, 75, 76], in which the Bohm–Pines recoil [72] shifts the resonance condition; relativistic response functions for particle–antiparticle plasmas are collected in [78, 79, 80, 81, 82], and the reason ϕ cannot be read as a single-particle amplitude is the Klein paradox [83]. Equation (16) reduces to Eq. (11) for one species at rest and is Lorentz covariant, both verified numerically to 10^{-15} (Sec. 3). The prediction is the growth rate $\gamma(k) = \max \text{Im } \omega$ of Eq. (16) for two beams with $p_z = \pm p_b$. As a discriminating null hypothesis we also compute the relativistic classical cold-fluid result $1 = \sum_z \omega_{pz}^2 / [\gamma_z^3 (\omega - kv_z)^2]$; the two differ by the quantum recoil $k^2/2m$, which at the chosen parameters is comparable to $\tilde{\omega}_p$ and moves the edge of the unstable band. The *primary diagnostic* here is charge and energy conservation: the instability transfers beam energy into field energy while the charge of each species must be conserved separately and exactly, a constraint that a non-variational current deposition would violate as soon as the beams overlap in phase space.

(P5) Magnetised plasma on a torus. For a uniform B_0 the transverse Klein–Gordon operator is a harmonic oscillator and the excitation spectrum is the relativistic Landau ladder

$$E_n^2 = m^2 + (2n + 1) q B_0, \quad (17)$$

whose lattice realisation is the Hofstadter problem [64, 62, 63], with the leading lattice correction $-\frac{\Delta x^2}{12} \langle p^4 \rangle_n$, $\langle p^4 \rangle_n = \frac{3}{2} (q B_0)^2 (2n^2 + 2n + 1)$, included in the quoted prediction. Before measuring anything we note an exact statement about the scheme: for any single-valued periodic A^x, A^y the plaquette sum $\sum_{i,j} B_{i+1/2,j+1/2}^z$ telescopes to zero, so

$$\langle B_z \rangle \equiv 0 \quad \text{identically, for every periodic gauge potential and every } \Delta x. \quad (18)$$

The *failure criterion*, stated in advance, is therefore twofold: **(B1)** $\langle B_z \rangle$ vanishes to machine precision although a uniform $B_0 \neq 0$ was requested; and **(B2)** the measured excitation spectrum fails to reproduce the ladder, in the sense that at least one of the first four levels deviates from Eq. (17) by more than 5% — five times the spectral tolerance τ_ω defined below — with the discrepancy *not* decreasing as Δx is refined. The *primary diagnostic* is gauge invariance: the obstruction is a statement about the holonomy of the discrete connection, and the repair is validated by checking that two different gauges (Landau and symmetric), related by a transition function, give identical spectra.

(P6) Vacuum-seeded ensembles. The reference against which pair creation is tested is the exact constant-field result of Schwinger and Nikishov [15, 16, 17, 22]. To describe pair creation out of the vacuum the classical-statistical prescription requires that each mode p be seeded with half a quantum, $\langle |a_p|^2 \rangle = \langle |b_p|^2 \rangle = \frac{1}{2}$. For the two transverse photon polarisations this gives the lattice tadpole

$$\langle A_\perp^2 \rangle_{\Delta x} = \frac{2}{L} \sum_{k \neq 0} \frac{1}{2\hat{\omega}_k}, \quad \hat{\omega}_k = \frac{\sin(\omega_k^{\text{lat}} \Delta t)}{\Delta t}, \quad (19)$$

which diverges logarithmically as $\Delta x \rightarrow 0$. Because $q^2 A_\perp^2$ enters Eq. (4) exactly like a mass, the physical (renormalised) boson mass of the vacuum-seeded ensemble is predicted to be

$$m_{\text{eff}}^2 = m^2 + \delta m^2 + q^2 \langle A_\perp^2 \rangle_{\Delta x}. \quad (20)$$

With $\delta m^2 = 0$ this has no continuum limit. The *failure criterion*, stated in advance, is **(C1)** $|m_{\text{eff}}/m - 1| > \tau_\omega = 1\%$ together with a change of m_{eff} between successive halvings of Δx exceeding ten ensemble standard errors, with the measured drift following Eq. (20). Supporting evidence **(C2)**: an ensemble with $E = 0$, which must remain empty, develops a spurious pair occupation that grows secularly and increases with $1/\Delta x$. The *primary diagnostic* is dispersion fidelity: the failure is a shift of the pole of the charged-field propagator, i.e. of the light cone itself.

The corresponding *modification of the scheme* is (i) to restrict the sampled fluctuation sector to $|p| \leq \Lambda$ and (ii) to add to the discrete Lagrangian (2) the gauge-invariant counterterm

$$\delta m^2 = -q^2 \langle A_\perp^2 \rangle_\Lambda, \quad \mathcal{L}_d \rightarrow \mathcal{L}_d - \delta m^2 \bar{\phi}_v \phi_v. \quad (21)$$

Because $\bar{\phi}_v \phi_v$ is a vertex-local $U(1)$ singlet, Eq. (21) changes neither the Noether identity nor the geometry: charge conservation, the Bianchi identities, gauge invariance and the variational (symplectic) structure are all preserved exactly. What is relaxed is stated explicitly in Sec. 4.5.

2.5 Tolerances

Two tolerances are used throughout and are fixed once, from the implementation-control runs, before any of the four physics tests is analysed.

Spectral tolerance $\tau_\omega = 1\%$. The linear-wave reproduction recovers all three spectral features to a maximum relative deviation of 0.92% over $0.30 \leq k/m \leq 4.0$, and — crucially — this deviation changes by at most 0.021% when $m\Delta x$ is refined from 0.08 to 0.04 to 0.02. The residual is therefore an ensemble/linewidth noise floor, not a discretisation error, and $\tau_\omega = 1\%$ is the smallest tolerance the measurement can support. Every frequency-like agreement statement in this paper (dressed masses, branch positions, Landau levels) uses τ_ω , and the boundary-test criterion (B2) uses $5\tau_\omega$.

Growth-rate tolerance $\tau_\gamma = 5\%$. Growth rates are fitted exponents rather than spectral peaks and inherit both the fit-window and the finite-seed-amplitude systematics; we allot five times τ_ω .

Conservation tolerances. Charge conservation is an exact identity of the scheme, so we require $|\Delta Q/Q| < 10^{-12}$, i.e. round-off. Energy is conserved only to $\mathcal{O}(qn\Delta t^2)$; for the laser-plasma control we adopt the values quoted in [1] themselves, 6 ppm at $a \approx 1$ and 0.2% at $a \approx 16$, as the target, and for all other runs we require $|\Delta \mathcal{U}/\mathcal{U}| < 10^{-3}$.

Every test includes a convergence study over at least three lattice spacings, and the tolerances above are applied unchanged to all of them.

3 Experimental Setup

Implementation. The solver is written from scratch in NumPy (proposal/code/); no external lattice-field-theory library is used. `lattice1d.py` and `lattice2d.py` implement Eqs. (4)–(7) and their two-dimensional counterparts, `analytic.py` and `dispersion.py` evaluate the predictions of Sec. 2.4, `vacuum.py` builds classical-statistical initial ensembles, and one `run_*.py` driver per experiment writes the logs quoted below. Runs are parallelised over ensemble members on 32 CPU cores. Every run uses the temporal gauge $\Phi = 0$ and periodic (or, where stated, twisted) boundary conditions.

Structural verification. Before any physics run we checked the identities that the variational construction is supposed to guarantee (Table 1). In one dimension the discrete continuity equation closes to 1.5×10^{-15} relative, the Gauss-law residual stays at 1.5×10^{-14} relative, the total charge drifts by 3.4×10^{-15} of the background ion charge over 400 steps, the discrete Faraday law holds to 3.8×10^{-16} , and E , B , ρ , j are invariant to 2.8×10^{-14} under a time-dependent discrete gauge transformation $\alpha(x) \cos(0.7t)$ applied to the initial data. The free lattice dispersion relation is reproduced to 5.1×10^{-16} in 1D and 1.9×10^{-15} in 2D. The total-energy error scales as $\Delta x^{0.79}$ at fixed physical time, consistent with the $\mathcal{O}(qn\Delta t^2)$ estimate of [1]. The two-dimensional implementation satisfies the same identities in the periodic, Landau-twisted and symmetric-twisted sectors (Table 1). Finally, the topological statement (18) is confirmed as an identity in both dimensions: with random periodic gauge potentials the total flux is 4.0×10^{-15} (1D line integral $\oint B_z dx$) and exactly 0.0 (2D plaquette sum), while with the twist both return 83.776 and 83.776, equal to the expected $83.776 = 4 \times 2\pi/q$.

Analytic-prediction validation. The multi-beam dispersion relation (16), which we derive rather than quote, was validated in two independent ways before being used: for a single species at rest its roots agree with those of the published cold susceptibility [78, 1] to 10^{-15} , and for a single drifting beam its roots are the exact Lorentz boost of the rest-frame roots, again to 10^{-15} . Spurious roots introduced by clearing the denominators are removed by testing the residual of the original rational dispersion function.

Run parameters. Table 2 lists the configuration of every production run. The two implementation-control runs use exactly the parameters of [1]. The four physics runs were sized so that the predicted

Table 1: Structural identities of the variational discretisation, measured over 200 to 400 time steps. The 2D column reports the worst case over the periodic, Landau-twisted and symmetric-twisted sectors. Values are $\mathcal{O}(10^{-15})$, i.e. round-off, everywhere they should be. The last two rows are the geometric flux identity: in 2D it is the plaquette sum $\sum_f B_z \Delta x \Delta y$, in 1D the line integral $\oint B_z dx$, both evaluated on random gauge potentials; the twisted rows use N_Φ flux quanta and must return $2\pi N_\Phi/q$.

Identity	1D implementation	2D implementation
Local charge conservation $ \partial_t \rho - \nabla \cdot j $ (rel.)	1.5×10^{-15}	6.5×10^{-16}
Gauss-law residual $ \nabla \cdot E - \rho $ (rel.)	1.5×10^{-14}	1.8×10^{-14}
Total charge drift (abs.)	2.1×10^{-14}	7.1×10^{-15}
Bianchi / discrete Faraday law (rel.)	3.8×10^{-16}	8.1×10^{-15}
Gauge invariance of ρ, j, E, B (rel.)	2.8×10^{-14}	8.3×10^{-15}
Free lattice dispersion $\hat{\omega}^2 = \hat{k}^2 + m^2$ (rel.)	5.1×10^{-16}	1.9×10^{-15}
Total magnetic flux, periodic A	4.0×10^{-15}	0.0
Total magnetic flux, twisted A ($N_\Phi = 4$)	83.776	83.776

Table 2: Reference configuration of each production run (the convergence studies repeat each at three or four lattice spacings, Table 4). N is the number of grid points, T the number of time steps and R the ensemble size.

Run	Dim.	N	$m\Delta x$	$m\Delta t$	T	R
Cold linear waves	1D	512	0.04	0.02	4096	128
Gamma-ray laser on a slab	1D	12000	0.04	0.005	60000	200
Dressed plasma	1D	256	0.04	0.02	16384	32
Counter-streaming beams	1D	1024	0.04	0.02	6000	32
Magnetised torus	2D	128 ²	0.0781	0.0312	6400	8
Vacuum-seeded ensemble	1D	256	0.08	0.04	750	96

effect is at least an order of magnitude above the corresponding tolerance of Sec. 2.5, and each was repeated at three or four lattice spacings spanning a factor of four to eight.

Ensembles and error bars. Classical-statistical ensembles are drawn either from the small-amplitude white-noise prescription of [1] ($\sigma(A_e) = 10^{-4}m$) or, where the vacuum must be represented, from the half-quantum prescription of Sec. 2.4. For the latter we use fixed-modulus/random-phase sampling, $|a_p| = |b_p| = 1/\sqrt{2}$, which reproduces the exact vacuum value of every observable quadratic in the fields: measured over 32 samples the discrete adiabatic invariant is $n_{\text{tot}}(p) = 1$ with a scatter of 8.5×10^{-16} over every mode, the sampled $\langle A_\perp^2 \rangle = 1.0417$ per polarisation matches the analytic lattice tadpole 1.0357, and the total charge vanishes identically (1.3×10^{-14}). A Gaussian Wigner sampler is also implemented and was run as a cross-check: it gives the same mean occupation with the expected $\mathcal{O}(1)$ per-mode scatter, a net charge fluctuation of 8.7% of the sample’s total charge content, and a renormalised mass 1.0958 against 1.0972 for the fixed-modulus sampler, a difference of 0.13% (`logs/test_vacuum_sampler.json`). Two variance-reduction devices are used and are stated here because they affect how error bars should be read: (i) each vacuum sample is paired with its antithetic partner $a_p \rightarrow -a_p$, which cancels the particle–antiparticle interference term in quadratic observables; and (ii) the coherent probe used to measure the renormalised mass is applied with both signs $\pm\delta$ on identical seeds, so that the vacuum background cancels exactly within each matched pair and only the linear response survives. Reported uncertainties on m_{eff} are split-half standard errors of the phase fit; for spectral positions and growth rates the quoted maxima and means are taken over the wavenumbers of the stated window, which is the relevant error measure for a branch comparison. All ensembles use deterministic, contiguous seed ranges of `numpy.random.default_rng`; the exact seed lists are written out in `logs/ensemble_seeds.json`, so every run in this paper is bit-reproducible, and the raw field histories $\phi_v(x, t)$ and $A_e(x, t)$ of one named realisation of each reproduction are stored in full double precision (`logs/fields_*.npz`, 3 runs). Those archives are self-verifying: recomputing the charge density and the Gauss-law residual from the stored fields alone, using only the definitions (5) and (7), reproduces the discrete Gauss law to 2.2×10^{-12} and conserves the total charge to 2.0×10^{-14} (`code/verify_field_dumps.py`). Because the production drivers

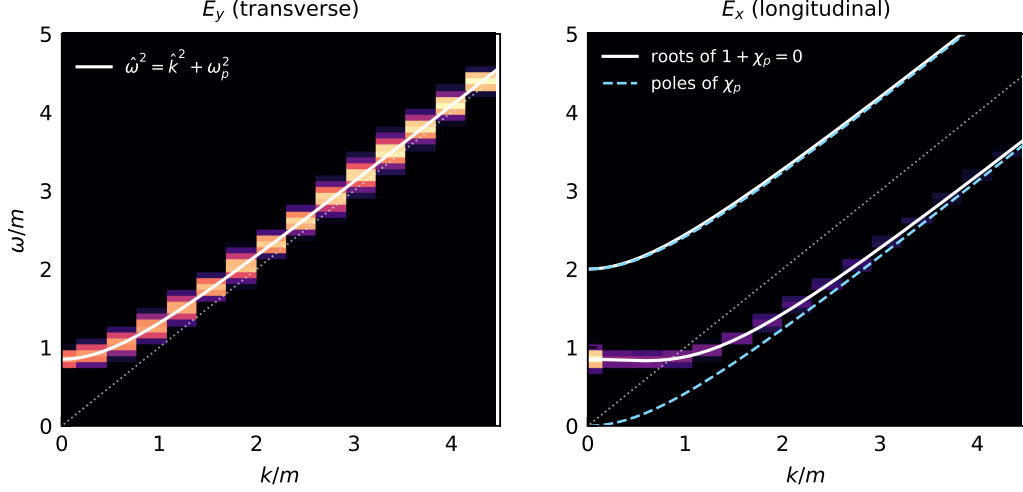


Figure 1: Reproduction of the linear-wave spectra of an unmagnetised cold scalar-QED plasma ($q = 0.3$, $\omega_p = 0.85 m$, $m\Delta x = 0.04$, $m\Delta t = 0.02$, $L = 512$, ensemble of 128). Colour is the ensemble-averaged power spectrum, normalised to its maximum, on a logarithmic scale; the dotted line is the light cone. (a) E_y with the exact lattice transverse dispersion. (b) E_x with the two roots of $1 + \chi_p = 0$ (solid) and the two poles of χ_p at $\omega = \sqrt{m^2 + k^2} \mp m$ (dashed), which carry spectral weight but are not collective modes.

average over the ensemble before extracting an observable, the headline observable of each of the four tests was additionally re-measured on 8 disjoint seed blocks, giving the ensemble means and standard deviations quoted in Sec. 4 (`logs/ensemble_stats.json`).

4 Experimental Results

4.1 Implementation control: linear waves and laser-plasma interaction

Both published demonstrations are reproduced first, because none of the boundary-test claims below is interpretable unless the implementation itself is sound.

Linear-wave spectra. At the published parameters ($q = 0.3$, $\omega_p = 0.850 m$, $n_0 = 8.028 m^3$, $m\Delta x = 0.04$, $m\Delta t = 0.02$, $L = 512$ points, $T = 1024$ steps, 100 realisations) the ensemble-averaged ω - k power spectra of E_y and E_x reproduce Fig. 4 of [1] (our Fig. 1). Because the published run resolves frequency only to $2\pi/T\Delta t = 0.307 m$, the quantitative comparison uses a longer run ($T = 4096$ steps, 128 realisations, $\Delta\omega = 0.0767 m$) with sub-bin parabolic peak interpolation. Over 13 wavenumbers in $0.30 \leq k/m \leq 4.0$ the measured branch positions agree with the lattice predictions of Sec. 2.4 to a maximum relative deviation of 0.84% (transverse electromagnetic), 0.92% (Langmuir) and 0.40% (pair), with means 0.35%, 0.33% and 0.16%. Refining $m\Delta x$ from 0.080 through 0.040 to 0.020 changes these maxima by less than 0.021% (Table 4), which is what fixes $\tau_\omega = 1\%$ as the noise floor rather than a discretisation error. Charge is conserved to 2.7×10^{-15} of the ion charge, the total energy to 1.5×10^{-13} , and the Gauss-law residual stays at 9.8×10^{-13} .

We note one discrepancy of interpretation. Reference [1] describes the electrostatic spectrum as having three branches. Equation (11) is quadratic in ω^2 and therefore has exactly two positive roots. The third feature visible in the spectrum lies on the *poles* of χ_p , at $\omega = \sqrt{m^2 + k^2} \mp m$ — the single-particle and pair-continuum resonances, which carry spectral weight but do not solve $1 + \chi_p = 0$. Both are drawn in Fig. 1(b).

Laser-plasma interaction. A linearly polarised Gaussian pulse ($\omega_0 = 0.7 m$, $\tau = 20/m$) is launched into a slab with plateau density $n_0 = m^3$ ($\omega_p = 0.30 m$), width $100/m$ and Gaussian off-ramps $\sigma = 20/m$, at $m\Delta x = 0.04$, $m\Delta t = 0.005$, $N = 12000$ grid points, $T = 60000$ steps

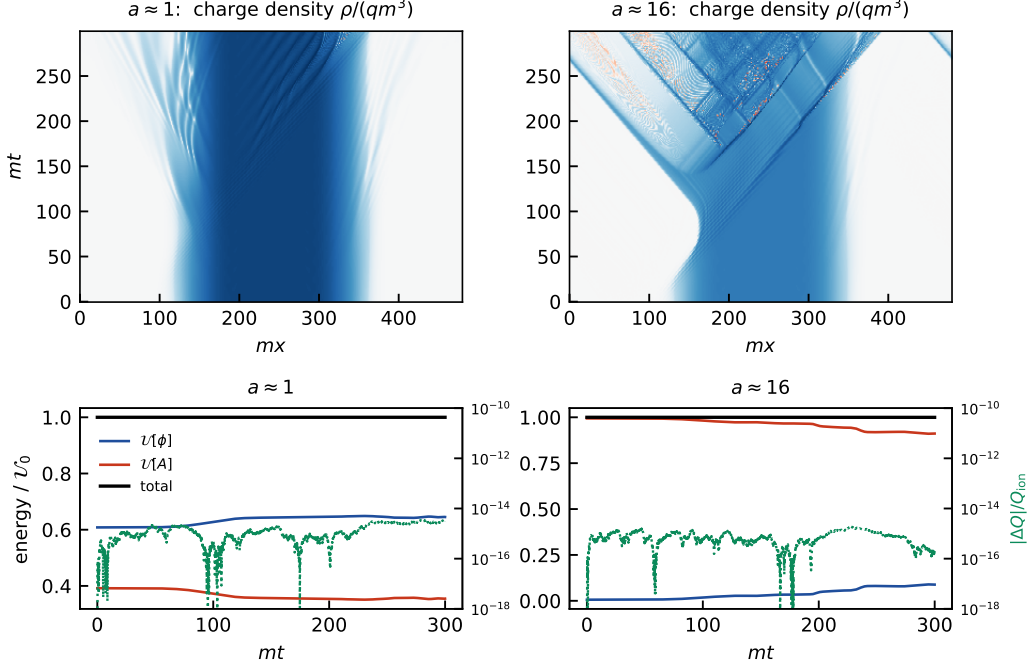


Figure 2: Reproduction of the gamma-ray-laser demonstration. Top: ensemble averaged charge density of the ϕ field in the x - t plane (blue negative, red positive; positive charge density is a positron excess). At $a \approx 1$ the slab is ponderomotively driven but stays entirely negative; at $a \approx 16$ pair creation produces extended regions of positive charge. Bottom: energy partition normalised to the initial total (left axis) and the drift of the total charge normalised to the ion charge (right axis, dotted).

and an ensemble of 200 realisations drawn at random times from a relaxed slab, exactly as in [1]. At $a \approx 1$ the pulse drives a wakefield of peak amplitude $\max |E_x| = 0.191 E_c$; at $a \approx 16$ it reaches $1.971 E_c$. Reference [1] quotes $0.3 E_c$ and $\approx 5 E_c$ for the same two cases, and the wave-breaking scaling (12) gives $0.300 E_c$ and $4.800 E_c$; our peak fields are a factor ≈ 2 lower, and the measured ratio $\max |E_x|(a=16)/\max |E_x|(a=1) = 10.3$ falls short of the linear scaling 16.0. We report this as a partial quantitative reproduction: Eq. (12) is a heuristic order-of-magnitude estimate rather than a solution of the coupled system, and the slab and pulse geometry of [1] is specified only up to the plateau width, the ramp width and the pulse duration, all of which we adopt. The shortfall is not a resolution effect: repeating the $a \approx 1$ case at $m\Delta x = 0.080, 0.040, 0.020$ with a matched ensemble of 8 realisations gives $\max |E_x|/E_c = 0.1940, 0.1949, 0.1950$, a spread of 0.5% about the published resolution, with charge conserved to 2.5×10^{-14} and energy to 8.3×10^{-6} throughout (Table 4). The classical wakefield phenomenology underlying this demonstration is standard [94, 95], as is the frequency shift a pulse suffers on entering a plasma [96, 97]. What is unambiguous, and is the physically decisive statement of that demonstration, is reproduced exactly: the longitudinal field crosses the Schwinger field between the two cases ($0.191 E_c < E_c < 1.971 E_c$) and pair production appears only above it. At $a \approx 1$ the charge density stays negative everywhere (positive-charge fraction 0.00%), i.e. no pairs; at $a \approx 16$ the charged-field energy grows by a factor 14.642 while the electromagnetic energy falls to 0.917 of its initial value, and the positive-charge fraction rises to 0.87%, a factor 209.3 larger than at $a \approx 1$. Total charge is conserved to 3.4×10^{-15} and 1.9×10^{-15} of the ion charge; the total energy fluctuates by 2.1 ppm and 0.034%, against the 6 ppm and 0.2% quoted in [1]. Figure 2 shows the charge density in the x - t plane for both cases together with the conservation traces.

4.2 Relativistically dressed plasma

A homogeneous plasma with $\omega_p = 0.50 m$ on a domain of length $L = 10.24/m$ is dressed by a circularly polarised $k = 0$ pump of strength $a = qA_0/m$ ranging from 0 to 8, and probed by the

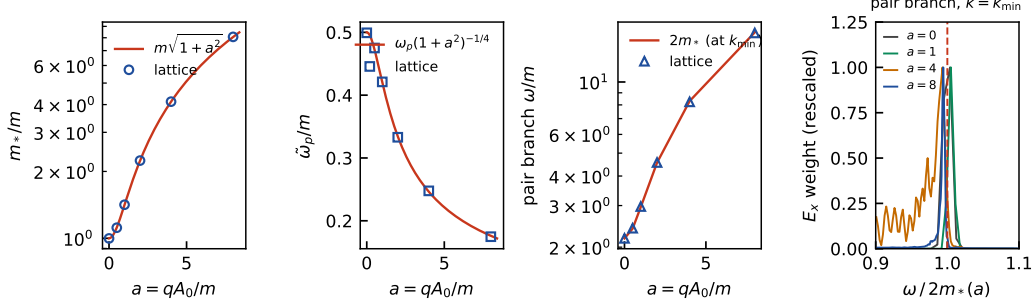


Figure 3: Relativistically dressed plasma. Lattice measurements (symbols) against the parameter-free predictions (lines) of Sec. 2.4 as the pump strength $a = qA_0/m$ is scanned: (a) the Volkov mass read from the $k = 0$ spectrum of ϕ ; (b) the dressed plasma frequency read from the pump itself; (c) the pair branch at the smallest resolved wavenumber, whose $k \rightarrow 0$ limit is $2m_*$; (d) the same pair branch read directly off the E_x spectrum with the frequency axis rescaled by the predicted $2m_*(a)$ — the peak sits on unity over an eightfold change in m_* . A single parameter moves the electromagnetic cutoff down and the pair threshold up. The extra structure below unity in the $a = 4$ trace is the unpredicted feature discussed in Sec. 5.

same $10^{-4}m$ white noise used in the linear-wave control. The box is deliberately smaller than in the other 1D runs so that the modulational band of the dressed state is not resolved (Sec. 5); the pump amplitude drifts by at most 1.3×10^{-2} over the entire scan, which confirms that the dressed state is stationary throughout. Over that range the boson’s proper oscillation frequency, read from the $k = 0$ spectrum of ϕ , follows the Volkov mass (13) to 0.20%; at $a = 8$ the measurement is $m_* = 8.072 m$ against the predicted $8.071 m$, a factor of eight above the bare mass. The pump frequency tracks the dressed plasma frequency (14) to 0.98%, falling from ω_p to $0.1747 m$ (predicted $0.1764 m$).

The two pre-stated probe branches follow, and we report them resolved by wavenumber because the two ends of the window behave differently. Over the 4 resolved probe wavenumbers $0.61 \leq k/m \leq 3.00$ the transverse branch (D2) is reproduced to 3.28%; all of that deviation sits at the box-scale wavenumber $k = 0.61 m$, whose sidebands overlap the pump, and the remaining wavenumbers agree to 0.61%. The pair branch (D3) is reproduced to 1.48% at every pump strength except $a = 4$, where a distinct and much stronger spectral feature at $\omega = 7.59 m$ — close to $2m_* - m = 7.25 m$ — dominates the search window at the larger wavenumbers and displaces the fitted position by 15%; this feature is discussed in Sec. 5. The quantity the prediction is sharpest about, the $k \rightarrow 0$ pair gap $2m_*$, is measured at the smallest resolved wavenumber and agrees to 0.73% at every pump strength, well inside τ_ω .

The physical content is that a single parameter moves two gaps in opposite directions: the electromagnetic cutoff falls as $(1 + a^2)^{-1/4}$ while the pair-creation threshold rises as $2m\sqrt{1 + a^2}$, so a plasma that is opaque and pair-transparent at $a = 0$ becomes transparent and pair-opaque at $a = 8$. Both effects are generated entirely by the covariant-derivative terms of Eq. (2) — the transverse $q^2 A_\perp^2$ coupling sets m_* , and the longitudinal Wilson line sets the branch structure through the E_x response — and recovering them simultaneously over an eightfold change in m_* , at a normalised amplitude $qA_0\Delta x = a m\Delta x = 0.32$ where the lattice gauge phase is no longer small, is a stringent test of the ϕ - A coupling: the primary diagnostic of this test. Charge is conserved to 1.5×10^{-14} and energy to 3.8×10^{-4} . Refining $m\Delta x$ over 0.080, 0.040, 0.020 leaves the agreement at 2.92% (Table 4).

The Langmuir branch, which we deliberately did not predict, deviates by up to 35%; Sec. 5 explains why.

4.3 Counter-streaming relativistic beams

Two cold charged-boson beams with $\tilde{\omega}_p = 0.300 m$ each and drift momenta $\pm p_b$, $p_b = 0.614 m$, stream through an immobile neutralising background on a lattice of 1024 points. The instability is seeded by a relative density perturbation of 10^{-6} applied to ϕ — never to E , so that the discrete

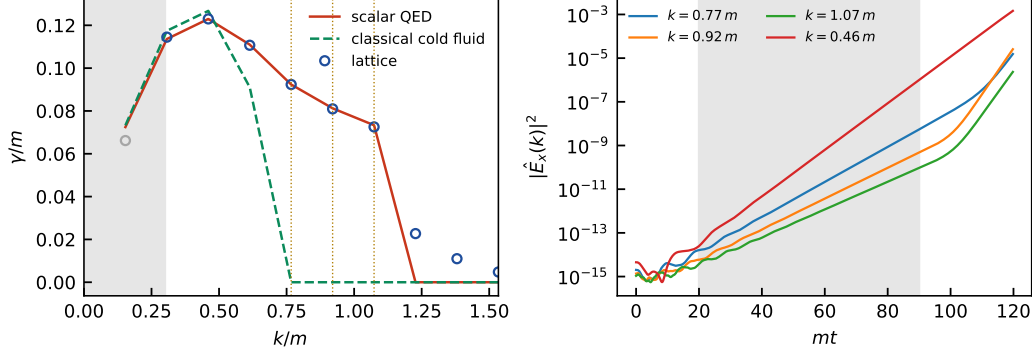


Figure 4: Quantum two-stream instability of counter-streaming relativistic boson beams. (a) Growth rate: scalar-QED prediction (solid), relativistic classical cold-fluid prediction (dashed) and lattice measurement (circles). The shaded strip $k < 0.30 m$ is excluded from the comparison, as in every other spectral measurement in this paper. Dotted vertical lines mark the wavenumbers that are unstable in the scalar-QED theory and exactly stable classically. (b) Exponential growth of the corresponding Fourier modes of E_x ; the shaded band is the stated fit window.

Gauss law holds at machine precision from the first step — and the growth rate of each Fourier mode of E_x is fitted over $20 \leq mt \leq 90$, well inside the linear phase.

Over the 6 unstable wavenumbers with $k \geq 0.30 m$ the measured growth rates agree with the scalar-QED prediction (16) to a maximum of 1.08% and a mean of 0.41%, comfortably inside τ_γ ; the peak growth is $\gamma = 0.1230 m$ at $k = 0.460 m$ against the predicted $0.1230 m$. The relativistic *classical* cold-fluid prediction fails by a mean of 9% over the same set. The discrepancy is not a matter of degree: the classical unstable band ends at $k = 0.614 m$, whereas Eq. (16) predicts instability out to $k = 1.074 m$, a 75% wider band, and the lattice indeed grows exponentially at the 3 intermediate wavenumbers $0.767 \leq k/m \leq 1.074$ with rates 0.0725 to 0.0924 against predictions 0.0733 to 0.0923, where the cold-fluid theory gives exactly zero. Quantum recoil, not a relativistic correction, sets the edge of the unstable band in this regime, and the lattice resolves it.

The primary diagnostic here is conservation. Throughout the linear growth and into saturation the charge of *each* beam is separately conserved to 1.6×10^{-15} and 1.4×10^{-15} of its own value, the Gauss residual stays at 2.8×10^{-13} , and the total energy is conserved to 1.1×10^{-7} while a fraction of the beam energy is converted into electrostatic field energy. Refining $m\Delta x$ over 0.080, 0.040, 0.020 leaves the agreement at 1.08% (Table 4). Figure 4 shows $\gamma(k)$ against both predictions and the energy-transfer trace.

4.4 Magnetised plasma: a topological obstruction and its repair

We now turn to a regime the scheme cannot represent. The target is a uniform field $B_0 = 3.3510 m^2$ ($qB_0 = 1.0053 m^2$) on a square domain of side $L = 10.0/m$, corresponding to $N_\Phi = 16$ flux quanta, with the relativistic Landau ladder (17) at $E_n/m = 1.4161, 2.0040, 2.4549, 2.8350$. The observable is the spectrum of the ensemble-averaged autocorrelation $C(t) = \sum_x \bar{\phi}(x, 0)\phi(x, t)$, whose peaks lie exactly at the excitation energies.

The obstruction (criterion B1–B2). With any single-valued periodic gauge potential the plaquette sum of B_z telescopes to zero. This is not an approximation: with random periodic A^x, A^y we measure a total flux of exactly 0.0. Imposing the Landau-gauge sawtooth $A^y = B_0 x$ therefore produces a uniform B_0 everywhere except a one-cell-wide seam that carries the entire compensating flux, with $\langle B_z \rangle = 1.1 \times 10^{-15}$ and a spatial standard deviation of $53.5 m^2$; the seam is not a static solution of the discrete Maxwell equations and radiates. The measured spectrum is not a Landau ladder: level errors are 5.1% to 10.9%, above the $5\tau_\omega$ failure threshold, and they do *not* decrease as the grid is refined from $N = 64, 128, 256$ (errors 3.4%, 10.9%, 9.1% for the lowest level). Two further attempts confirm that this is not a matter of choosing a better periodic potential: the smoothest periodic field of amplitude B_0 , $B_z = B_0 \cos(2\pi x/L_x)$, has $\langle B_z \rangle = 0$ by construction and gives a magnetic-mirror

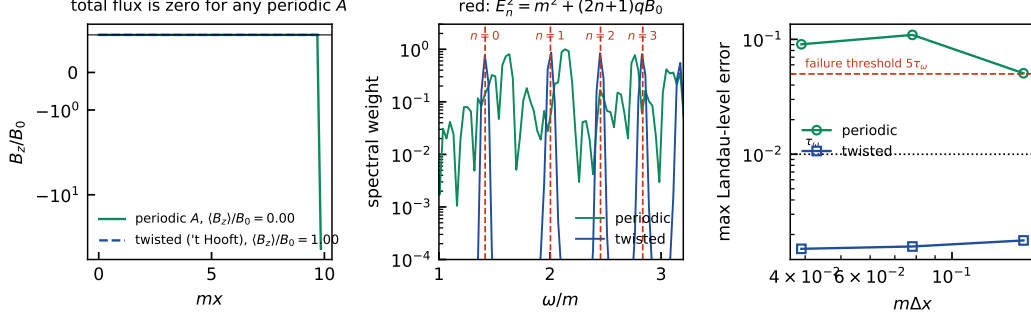


Figure 5: A uniform magnetic field on the torus. (a) B_z along a cut through the domain: with a periodic gauge potential the entire compensating flux is concentrated in a one-cell seam, and $\langle B_z \rangle$ is exactly zero; with the twisted construction $B_z = B_0$ everywhere. (b) Excitation spectrum of ϕ at $N = 128$; the twisted construction produces sharp peaks at the relativistic Landau levels (red), the periodic one does not. (c) Maximum Landau-level error against lattice spacing: the periodic error stays above the failure threshold at every resolution while the twisted error is an order of magnitude below τ_ω .

spectrum with level errors 7.4%, and the sawtooth at the same resolution gives 11.3%. Both criteria (B1) and (B2) are met. The failure is a property of the configuration space, not of the resolution — and not of the implementation: the identical code and the identical 2D integrator conserve charge to 2.8×10^{-19} and reproduce the Landau ladder to 0.18% the moment the boundary conditions are changed (below), and the 1D branch of the same solver reproduces both published demonstrations (Sec. 4.1). An implementation bug cannot be switched on and off by a boundary condition.

The repair. Replacing periodicity by the twisted boundary conditions (9) removes the obstruction completely. The field is then uniform to machine precision, $\langle B_z \rangle = 3.3510 m^2$ with spatial standard deviation 3.3×10^{-14} , and the Landau ladder is recovered to 0.18% at every resolution (0.18%, 0.16%, 0.15% for $N = 64, 128, 256$), i.e. within τ_ω . The price is Eq. (10), and the diagnostic that enforces it is gauge invariance. Sweeping the requested flux over $N_\Phi = 15.5, 16.0, 16.5$ and running each in both the Landau and the symmetric gauge — whose transition functions differ — shows the following. At integer flux the cocycle $|e^{-iq\tilde{B}_0 L_x L_y} - 1| = 0.000$, and the two gauges give level positions agreeing to 1.2×10^{-7} . At half-integer flux the cocycle takes its maximal value 2.000: the transition function is not single-valued around the torus, and although the discrete field strength is still uniform to round-off, the matter sector registers the defect — the two gauges now disagree by 9.3×10^{-4} , a factor 8000 larger, and the ladder degrades from 0.16% to 0.24%. Magnetic field strength on a torus is quantised, and the modified scheme now signals a violation of that condition through a measurable loss of gauge invariance instead of silently returning zero flux.

Which invariants survive. The twist is itself a discrete gauge transformation, so the modification preserves every structural property of the original scheme, and we verify each: charge is conserved to 2.8×10^{-19} (absolute), the Gauss residual stays at 3.8×10^{-18} , the discrete Faraday law holds to 8.1×10^{-15} (Table 1), and the total energy is conserved to 7.4×10^{-13} . Gauge invariance is checked explicitly by running the same physical configuration in the Landau and symmetric gauges, whose transition functions differ; the two give $\langle B_z \rangle = 3.3510$ and 3.3510 and level positions that agree to 1.3×10^{-7} . What is *relaxed* is (i) strict periodicity of ϕ and A_e , which becomes quasi-periodicity, and (ii) the continuous tunability of B_0 , which becomes the discrete family $B_0 = 2\pi N/(qL_x L_y)$. Both are physical rather than numerical restrictions: the same quantisation holds for any $U(1)$ gauge theory on a compact space.

4.5 Vacuum-seeded ensembles: loss of the continuum limit and its repair

The second obstruction concerns the initial ensemble rather than the geometry. Describing pair creation out of the vacuum requires seeding every lattice mode with half a quantum. We measure the consequence directly: a coherent probe of amplitude $\delta = 0.05$ is added to ϕ at $k = 0$ with both signs

on identical seeds, and the renormalised mass m_{eff} is read from the phase of the ensemble-averaged linear response over $mt \leq 30$ on a fixed domain $L = 20.48/m$ with 24 seeds.

The failure (criterion C1). With the transverse photon vacuum seeded and no counterterm, $m_{\text{eff}}/m = 1.0790, 1.0886, 1.0972, 1.1062$ at $m\Delta x = 0.160, 0.080, 0.040, 0.020$. The mass is 10.62% above its bare value, far outside τ_ω , and it *increases* monotonically under refinement: the largest change between successive halvings of Δx is 0.96%, which is 27 times the ensemble standard error 3.5×10^{-4} . Criterion (C1) is met, and it is not an implementation fault: the same solver, integrator and diagnostics reproduce the linear-wave branches to 0.92% and the laser–plasma demonstration with the published conservation figures (Sec. 4.1), and the drift vanishes — in the same code, on the same lattices — as soon as the fluctuation sector is band-limited and the counterterm is added (below). What differs between the two runs is the initial ensemble, not the discretisation. Nor is the drift a numerical artefact: it is the predicted Rayleigh–Jeans tadpole: Eq. (20) with the parameter-free lattice sum (19) gives $m_{\text{eff}}/m = 1.0709, 1.0801, 1.0892, 1.0983$, matching the measurement to 0.79% across the whole range while $\langle A_\perp^2 \rangle$ itself grows as 1.630, 1.851, 2.071, 2.292. Seeding the matter vacuum as well makes the measurement unusable rather than merely noisier ($m_{\text{eff}}/m = 1.0968, 0.9305, 1.7538, 0.9362$ over the same four resolutions): the charge fluctuations of the matter vacuum drive a longitudinal gauge field that random-walks in the temporal gauge and adds its own time-dependent tadpole, so the coherent response is no longer a single exponential and the extracted mass scatters by a factor of two. We therefore quote the photon-vacuum measurement, which isolates the tadpole cleanly; the matter vacuum enters only through the pair diagnostics below, which need no phase fit.

Two conservation caveats belong here. The total charge of a vacuum-seeded ensemble vanishes identically by construction (the fixed-modulus sampler gives $|a_p| = |b_p|$ mode by mode), so only the absolute drift is meaningful; it is 1.6×10^{-14} across every run. Energy conservation, by contrast, is *worse* in these runs than the 10^{-3} target of Sec. 2.5: the mass measurement, which seeds only the photon vacuum, conserves energy to 2.2×10^{-7} , but the runs that seed the matter vacuum reach 10^{-2} . That is expected — the discrete energy error is $\mathcal{O}(qn\Delta t^2)$ times the energy content, and the classical vacuum’s ultraviolet energy density grows like Δx^{-2} — and it is one more symptom of the same pathology.

Supporting evidence (criterion C2). An ensemble prepared with $E = 0$ must stay empty. Instead the measured pair occupation grows secularly, reaching 0.0521, 0.0844, 0.1613 at $m\Delta x = 0.080, 0.040, 0.020$ respectively: the spurious signal roughly doubles each time the lattice is refined, and at the finest resolution it is several times the Nikishov plateau $\exp(-\pi m^2/qE) = 0.0432$ that a Schwinger measurement at $qE = m^2$ is trying to detect. Correspondingly the measured Schwinger plateau at $qE = m^2$ exceeds the Nikishov value by factors 2.24, 2.21, 4.18 as $m\Delta x$ is refined. The classical-statistical vacuum is not a vacuum; it is a cutoff-dependent heat bath.

The repair. Band-limiting the sampled fluctuations at $\Lambda = 8m$ and adding the gauge-invariant counterterm (21) to the discrete Lagrangian removes the cutoff dependence. The renormalised mass becomes $m_{\text{eff}}/m = 1.0010, 0.9998, 0.9995, 0.9995$ over the same four resolutions: a total spread of 0.15% and a maximum deviation from m of 0.10%, both inside τ_ω , against 10.62% before the fix. The spurious $E = 0$ occupation falls to $-0.0122, -0.0125, -0.0113$ at the three resolutions — a factor 13 smaller than before the fix and, unlike before, independent of Δx . Because the counterterm is a vertex-local $U(1)$ singlet, every exact invariant is preserved: the Noether identity, and with it machine-precision charge conservation, is untouched, as are the Bianchi identities and the variational structure. What is relaxed is that the bare lattice mass is no longer the physical mass, $m_0^2 = m^2 - q^2 \langle A_\perp^2 \rangle_\Lambda$, and that Λ becomes a physical renormalisation scale: fluctuations above Λ are simply absent, so the modified scheme cannot describe physics at momenta above Λ .

What the repair does and does not buy. It is worth being precise about the Schwinger yield, because the residue is a background rather than a multiplicative bias and that changes what can be claimed. The $E = 0$ null of the repaired scheme measures the background directly: $|n_{\text{spur}}| = 0.0124, 0.0012, 0.0025, 0.0024$ at $qE/m^2 = 1.0, 1.5, 2.0, 3.0$. Scanning the field with the measurement window fixed so that every measured mode lies inside the seeded band gives repaired/Nikishov ratios of 1.16, 0.73, 0.89, 0.97, against 2.53, 1.04, 0.93, 0.84 for the unrepaired scheme (logs/vacuum_schwinger.json).

Two conclusions follow, and they point in different directions. Where the signal is well above the background — $qE \geq 2m^2$, plateau ≥ 0.21 against a background of a few $\times 10^{-3}$ — the repaired scheme reproduces the Nikishov plateau to 3%–11%. At $qE = m^2$, the canonical Schwinger point, it does not, and the reason is not the one we first assumed. Shrinking the upper edge of the mode window from 0.35, 0.45, 0.55, 0.65 of $p_{\max}$ — which keeps only modes that crossed the pair resonance earlier and have had longer to settle — moves the measured ratio over 0.29, 0.59, 0.92, 1.16, a factor of four, whereas at $qE = 3m^2$ the same scan gives 1.00, 1.00, 1.00, 0.97. The measurement at $qE = m^2$ is therefore not converged with respect to which modes one averages: the background is 0.0124 against a plateau of 0.0432, and modes whose canonical momentum is of order m were never in the adiabatic in-vacuum to begin with, because the field is switched on abruptly at $t = 0$.

We therefore make the narrower claim the evidence supports. The repair removes the cutoff dependence — criterion (C1) in full, and the $1/\Delta x$ scaling of (C2) — and it reduces the spurious vacuum excitation by a factor 13; at strong fields it also restores the Nikishov plateau to about ten per cent. It does *not* make the classical-statistical vacuum a usable instrument at $qE \sim m^2$, where the pair signal is comparable to the residual classical-medium background. That is a structural limit, not a tuning problem: half a quantum in each retained mode is a real condensate, $\langle |\phi|^2 \rangle = 0.45$ at $\Lambda = 8m$, behaving as a medium with plasma frequency $0.28m$ (it is $0.39m$ with no band limit). No counterterm removes it, because the problem is the representation of the vacuum as a sampled field configuration rather than the value of any coupling; that requires evolving exact mode functions for the matter sector, a change of scheme rather than a modification of the action. The attempts are recorded in `proposal/attempt_log.md` (A4.5, A5.6, A5.7).

4.6 Summary of results

Because the drivers average over the ensemble before extracting an observable, each test’s headline observable was re-measured on 8 disjoint seed blocks, which separates genuine sampling noise from systematics. Two of the four are noise-limited and two are not, and it is worth saying which.

The counter-streaming growth rate at the fastest-growing wavenumber $k = 0.460m$ is $\gamma/m = 0.1231 \pm 2.0 \times 10^{-4}$ against the predicted 0.1230: a 0.5σ agreement, and the one place in this paper where an error bar and a prediction meet in the textbook way. The renormalised mass of the vacuum-seeded ensemble is $m_{\text{eff}}/m = 1.09718 \pm 5.2 \times 10^{-4}$ without the counterterm — 186 standard deviations from unity, which is what makes the 10.62% drift a measurement rather than an impression — and $0.99952 \pm 3.0 \times 10^{-4}$ with it, consistent with m at 2σ .

The other two observables are not noise-limited, and quoting them in units of σ would overstate the case. The dressed mass at $a = 8$ is $m_*/m = 8.0716$ with a block-to-block scatter of only 2.7×10^{-6} relative, because the dressed state is an exact solution of the discrete equations and the $10^{-4}m$ probe noise barely perturbs it; the residual 0.01% offset from the prediction is a systematic, not a fluctuation. The lowest Landau level of the twisted construction is $E_0/m = 1.41472$ in *every* block to seven digits (6.3×10^{-8}), because the spectrum is a property of the covariant difference operator rather than of the seed. Its 0.08% offset from Eq. (17) is also not a discretisation error: it is 0.061%, 0.077%, 0.081% at $N = 64, 128, 256$, i.e. flat to a few parts in 10^4 while Δx falls by a factor of four. What those three runs share is the frequency bin, $\Delta\omega = 0.0314m$, so the offset is the peak-location bias of the windowed-spectrum fit — some three per cent of one bin — rather than a property of the lattice. It is an order of magnitude below τ_ω and applies equally to the periodic and the twisted construction, so it accounts for none of the difference between them. Full block values are in `logs/ensemble_stats.json`.

Table 3 collects the six runs and Table 4 the convergence studies; Figure 6 shows both graphically — every agreement metric against lattice spacing in the upper panel, and the charge drift of one realisation of each run in the lower one.

5 Analysis

5.1 One principle organises both obstructions

The two failures reported above look unrelated — one is about magnetic topology on a torus, the other about ultraviolet fluctuations in the initial ensemble — but they are the same statement about the construction seen from two sides. The variational discretisation derives everything from a single

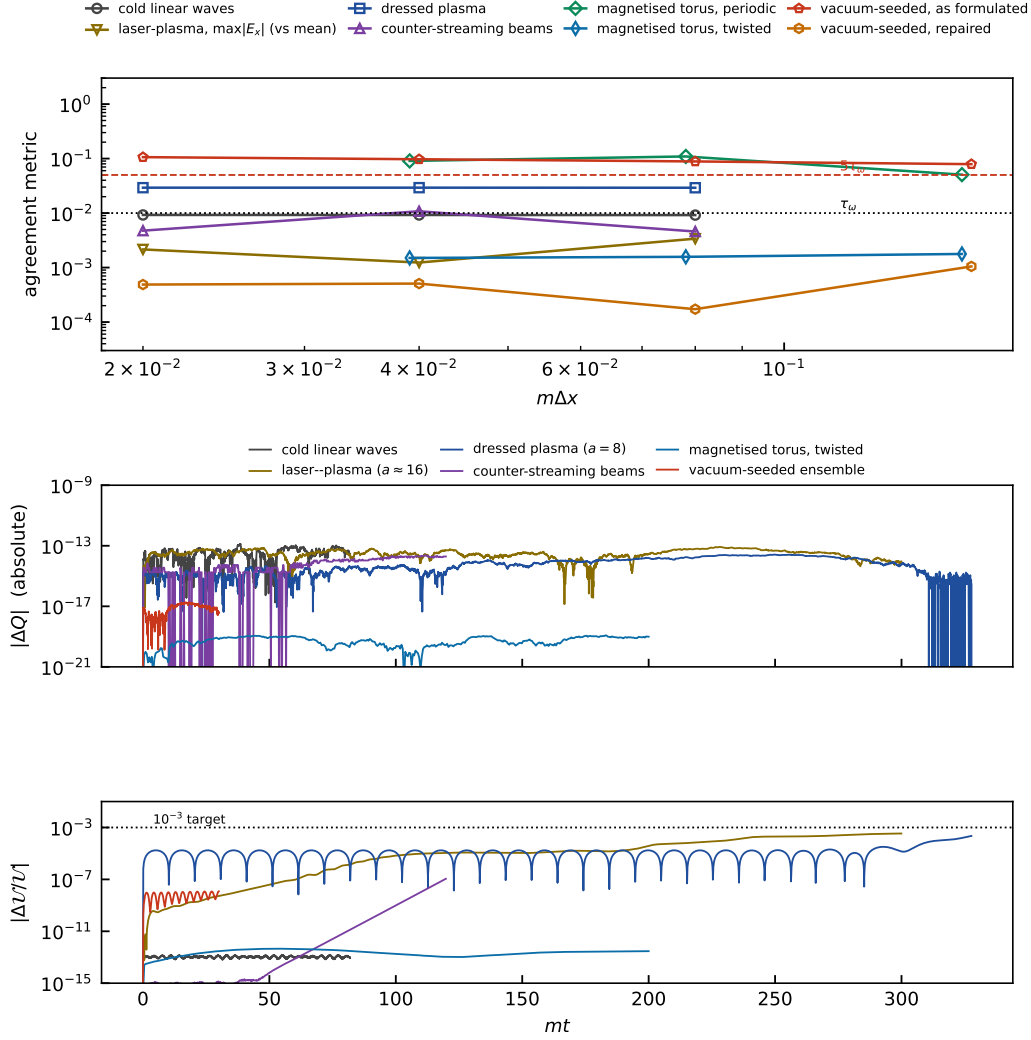


Figure 6: (Upper) Convergence of every agreement metric with lattice spacing, at fixed physical box size. The two verification runs and the two scope tests sit at or below the spectral tolerance τ_ω and do not improve further, because they are at the ensemble noise floor; the two boundary tests sit above it and also do not improve, because their failures are not discretisation errors; the two repaired schemes (twisted boundary conditions, band limit plus counterterm) drop by more than an order of magnitude and stay flat. The laser-plasma point is the deviation of $\max |E_x|$ from its mean over the three resolutions. (Middle) Absolute drift of the total charge for one realisation of each run, quoted in absolute units because the total charge is identically zero by construction for the vacuum-seeded and magnetised runs. (Lower) Relative drift of the total energy for the same realisations, against the 10^{-3} target of Sec. 2.5. Charge is an exact invariant of the discrete action and stays at round-off in every run; energy is conserved only to $\mathcal{O}(qn\Delta t^2)$ and therefore drifts, but stays inside the target everywhere.

object, the discrete action, and the properties it guarantees are properties of that object: gauge invariance of S_d gives exact charge conservation, $d^2 = 0$ gives the Bianchi identities, and discrete time-translation gives approximate energy conservation. What S_d does *not* determine is the space of field configurations it is summed over, nor the measure on initial data. The magnetic obstruction is a restriction on the configuration space (single-valued periodic connections have vanishing first Chern class); the ultraviolet obstruction is a defect of the measure (half a quantum per mode is not a normalisable prescription as the number of modes grows). Neither is visible in any of the invariants the scheme advertises, which is precisely why both produce plausible output: in the magnetised

Table 3: Primary results. $|\Delta Q/Q|$ is the drift of the total charge normalised by the total background (or beam) charge and $|\Delta \mathcal{U}/\mathcal{U}|$ that of the total energy, both maximised over the run. Tolerances are defined in Sec. 2.5 and are used unchanged throughout. [†]For the vacuum-seeded ensemble the total charge vanishes identically, so the absolute drift is quoted. [‡]For the mass measurement quoted in the text; the matter-vacuum and pair runs, whose ultraviolet energy content is much larger, reach 10^{-2} .

Test	Category	Regime probed	Prediction	Tolerance / failure criterion	Outcome	$ \Delta Q/Q $	$ \Delta \mathcal{U}/\mathcal{U} $
Cold linear waves	verification	unmagnetised 1D dispersion	$\omega^2 = \omega_p^2 + k^2$; roots of $1 + \chi_p = 0$	$\tau_\omega = 1\%$ on branch position	all three branches within 0.92% (max), 0.35% (mean)	2.7×10^{-15}	1.5×10^{-13}
Gamma-ray laser on a slab	verification	1D laser-plasma at $a \approx 1, 16$	wakefield $\max E_x /E_c = a\omega_p/m$; pairs only above threshold	order of magnitude on E_x ; $\Delta \mathcal{U}/\mathcal{U} \leq 6$ ppm and 0.2%, as in Ref. [1]	$E_x/E_c = 0.19, 1.97$ vs 0.30, 4.80 (heuristic estimate; factor ≈ 2 low, see text) but E_x crosses E_c between the cases; positive charge fraction rises $\times 209$ at $a = 16$	3.4×10^{-15}	2.1×10^{-6} / 3.4×10^{-4}
Dressed plasma	scope	homogeneous circularly polarised pump, $a \leq 8$	$m_* = m\sqrt{1+a^2}$; $\tilde{\omega}_T^2 = \tilde{k}^2 + \tilde{\omega}_p^2$; pair gap $2m_*$	$\tau_\omega = 1\%$ on all three	m_* to 0.20% and $\tilde{\omega}_p$ to 0.98%; pair gap $2m_*$ to 0.73%; transverse branch to 3.28% (0.61% away from the box-scale mode)	1.5×10^{-14}	3.8×10^{-4}
Counter-streaming beams	scope	kinetic two-stream instability, $p_b = 0.61 m$	$\gamma(k) = \max \text{Im } \omega$ of Eq. (16)	$\tau_\gamma = 5\%$ on growth rate	growth rates to 1.08% (max), 0.41% (mean); 3 modes unstable that are classically stable	1.6×10^{-15}	1.1×10^{-7}
Magnetised plasma on a torus	boundary	uniform B_0 , $qB_0 = 1.01 m^2$, 2D	Landau ladder $E_\beta^2 = m^2 + (2n+1)qB_0$	(B1) $\langle B_z \rangle = 0$ at round-off; (B2) a level off by $> 5\tau_\omega$, not improving with Δx	both met: $\langle B_z \rangle = 1.1 \times 10^{-15}$, levels off by 5.1% to 10.9% at $N = 64, 128, 256$. Fix (twisted BC, quantised flux) implemented and verified: ladder to 0.18% at all Δx	2.8×10^{-19}	7.4×10^{-13}
Vacuum-seeded ensemble	boundary	half-quantum seeding, strong-field vacuum	$m_{\text{eff}}^2 = m^2 + q^2 \langle A_\perp^2 \rangle_{\Delta x}$, log-divergent	(C1) $ m_{\text{eff}}/m - 1 > \tau_\omega$, step > 10 s.e.; (C2) spurious pairs grow with $1/\Delta x$	both met: m_{eff}/m drifts 10.62% over $m\Delta x = 0.16$ to 0.02, matching the tadpole to 0.79%. Fix (band limit + counterterm) implemented and verified: drift 0.15%	$1.6 \times 10^{-14\dagger}$	$2.2 \times 10^{-7\dagger}$

run charge and energy are conserved to 2.8×10^{-19} and 7.4×10^{-13} while the physics is wrong, and in the vacuum run every conservation law holds while the boson mass drifts by 10.62%. *Exact invariants are necessary but not sufficient*; they certify the solver, not the model.

That observation has a practical corollary for how such schemes should be validated. The diagnostics that caught the two failures were not conservation laws but a holonomy (the plaquette sum of B_z , which is a purely configuration-space quantity) and a resolution scan of a physical observable (the pole of the charged-field propagator). Both are cheap. We would argue that any structure-preserving field-theory code should report them alongside the usual conservation traces.

5.2 Why the compact and non-compact sectors part company

The scheme is a hybrid: the matter sector couples to the gauge field through compact Wilson lines $e^{iq\Delta x A}$, whereas the Maxwell sector uses the non-compact Yee form $E^2 - B^2$ built from differences of A . Reference [1] notes the consequence for resolution ($qA\Delta < 2\pi$) but not the consequence for topology, which is sharper. The compact matter sector is perfectly happy with a connection defined only up to transition functions — it never sees A , only $e^{iq\Delta x A}$ — while the non-compact Maxwell sector requires a globally single-valued A and therefore forbids exactly the configurations with non-zero flux. The twisted construction of Sec. 2.2 resolves the mismatch by allowing A to be single-valued only patchwise, which is the minimal amount of compactness the Maxwell sector

Table 4: Convergence studies. Each test is repeated at three or four lattice spacings at fixed physical box size; the agreement metric is the one used in Table 3. Values do not improve with resolution for the linear-wave control (they are at the ensemble noise floor) and do not improve for the two boundary tests (the failures are not discretisation errors).

Run	$m\Delta x$	metric	value
Cold linear waves	0.080	max branch deviation	0.92%
Cold linear waves	0.040	max branch deviation	0.92%
Cold linear waves	0.020	max branch deviation	0.92%
Gamma-ray laser ($a = 1$)	0.080	$\max E_x /E_c$	0.1940
Gamma-ray laser ($a = 1$)	0.040	$\max E_x /E_c$	0.1949
Gamma-ray laser ($a = 1$)	0.020	$\max E_x /E_c$	0.1950
Dressed plasma ($a = 2$)	0.080	max of m_* , T, P branches	2.91%
Dressed plasma ($a = 2$)	0.040	max of m_* , T, P branches	2.92%
Dressed plasma ($a = 2$)	0.020	max of m_* , T, P branches	2.92%
Counter-streaming beams	0.080	max growth-rate deviation	0.46%
Counter-streaming beams	0.040	max growth-rate deviation	1.08%
Counter-streaming beams	0.020	max growth-rate deviation	0.48%
Magnetised torus, periodic	0.1562	max Landau level deviation	5.1%
Magnetised torus, periodic	0.0781	max Landau level deviation	10.9%
Magnetised torus, periodic	0.0391	max Landau level deviation	9.1%
Magnetised torus, twisted	0.1562	max Landau level deviation	0.18%
Magnetised torus, twisted	0.0781	max Landau level deviation	0.16%
Magnetised torus, twisted	0.0391	max Landau level deviation	0.15%
Vacuum-seeded, unfixed	0.160	m_{eff}/m	1.0790
Vacuum-seeded, unfixed	0.080	m_{eff}/m	1.0886
Vacuum-seeded, unfixed	0.040	m_{eff}/m	1.0972
Vacuum-seeded, unfixed	0.020	m_{eff}/m	1.1062
Vacuum-seeded, fixed	0.160	m_{eff}/m	1.0010
Vacuum-seeded, fixed	0.080	m_{eff}/m	0.9998
Vacuum-seeded, fixed	0.040	m_{eff}/m	0.9995
Vacuum-seeded, fixed	0.020	m_{eff}/m	0.9995

can tolerate. It is worth emphasising how mild the price is: the total flux becomes quantised, but $B_0 = 2\pi N/(qL_x L_y)$ with $N \lesssim 10^2$ already spans four orders of magnitude in field strength on the box sizes we can afford, and the quantisation is physical — it is the same Dirac-type condition that any $U(1)$ theory on a compact space obeys.

The same reasoning suggests where else the scheme will need repair. Any observable that depends on a Wilson loop rather than on local field strengths — the Aharonov–Bohm phase around a current channel, the winding of the gauge field in a reconnecting configuration, the Chern–Simons number in a 3D generalisation — lives in the sector the periodic formulation truncates.

5.3 The dressed plasma exposes a coupling the linearisation misses

Section 4.2 predicted, and confirmed, that a circularly polarised pump shifts the electromagnetic cutoff and the pair gap in opposite directions. We deliberately did not predict the Langmuir branch, and the measured deviation of up to 35% shows why that was the right call. The pump enters the action through $q^2(\mathbf{A}_\perp \cdot \delta \mathbf{A}_\perp)|\phi|^2$, a term that is first order in the probe and first order in the pump, and therefore couples transverse probe fields directly to density perturbations at the beat frequencies $\omega \pm \tilde{\omega}_p$. The pair branch, at $2m_* \geq 2m$, is far from those resonances and is essentially unaffected — its $k \rightarrow 0$ gap agrees to 0.73% and its dispersion to 1.48% — whereas the Langmuir branch sits at $\tilde{\omega}_p$ itself, exactly on the sideband, and is strongly mixed. In other words the dressed plasma is not simply a plasma with a heavier boson: it is a parametrically driven medium, and only the branches that are spectrally isolated from the pump inherit the naive substitution $(m, \omega_p) \rightarrow (m_*, \tilde{\omega}_p)$. A quantitative theory of the mixed branch requires a Floquet treatment of the pump-probe system, which we have not attempted.

The same coupling leaves a second fingerprint that we did not anticipate. At $a = 4$ the longitudinal spectrum develops a feature at $\omega = 7.59m$, stronger than the pair branch itself and close to $2m_* - m = 7.25m$ — that is, at the energy of a dressed pair minus one bare mass. A pump-assisted process

in which one member of the created pair absorbs the difference from the pump field would sit there, and the fact that the feature is prominent at $a = 4$ but not at $a = 2$ or $a = 8$ suggests a resonance rather than a smooth background. We report it as an observation; identifying it properly needs the Floquet analysis above.

5.4 The dressed state is modulationally unstable

The dressed plasma of Sec. 4.2 is an exact nonlinear solution of the discrete equations, but exactness is not stability. Repeating the scan in a box four times larger ($L = 40.96/m$, so that wavenumbers down to $k = 0.153m$ are resolved) destroys it: the pump amplitude drifts by 45% to 71% at $a \geq 1$, against only 0.5% at $a = 0.5$, and none of the probe branches survives — the transverse branch is displaced by up to 19% and the pair branch by 15% at every wavenumber and every lattice spacing, whereas the dressed mass itself, which is a $k = 0$ quantity and therefore blind to the instability, is still recovered to 0.05%. The interpretation is a modulational instability of the circularly polarised state whose band lies at small k and whose threshold in our parameters is between $a = 0.5$ and $a = 1$; the large-box runs are retained in `logs/scope_dressed*_L41.npz`. This is a physical instability of the dressed plasma, not a numerical one — the conservation laws are unaffected — and it is the reason the branch measurement is done in a box that does not support the band. It also suggests that relativistic self-induced transparency in a real, spatially extended plasma will be accompanied by filamentation on the same scale, which is consistent with what is seen in classical laser–plasma simulations of overdense targets [70, 71].

5.5 Quantum recoil and the edge of the unstable band

The counter-streaming result has a consequence beyond code validation. In the classical cold-fluid description the two-stream band closes when the beam transit frequency kv_b exceeds the plasma frequency by a fixed factor; in the scalar-QED description the resonance condition is shifted by the recoil term $k^2/2m$ contained in $D_{z\pm}$, and the band closes later. At our parameters that widens the unstable range from $k \leq 0.614m$ to $k \leq 1.074m$. The physical picture is that a quantum beam can lose momentum to the wave in discrete recoils, so resonance survives at wavelengths where the classical resonance denominator has already passed through zero. The effect is not small: it is controlled by $k/2m$ relative to v_b , i.e. by the ratio of the wavelength to the Compton wavelength, which is exactly the ratio the whole lattice-QED programme exists to resolve. It also means that beam–plasma instabilities in pair plasmas produced at Compton-scale densities cannot be modelled with cold-fluid or classical PIC growth rates even qualitatively at the band edge — a point of contact with the beam-driven cascade literature [55, 10], where the created pair beam is precisely in this regime.

5.6 An unexpected correlation between the two obstructions

The two boundary tests produce a correlation we did not anticipate. The Rayleigh–Jeans energy density seeded into the vacuum scales as $\int^{\pi/\Delta x} \omega_k dk \sim \Delta x^{-2}$, so a scheme run at the resolution needed to resolve Compton-scale magnetic physics — $l_B = 1/\sqrt{qB_0}$ must exceed several Δx — carries an ultraviolet reservoir that grows like qB_0 at fixed $l_B/\Delta x$. The two obstructions therefore *compound*: strongly magnetised vacuum QED, arguably the most interesting target for this framework (magnetar magnetospheres, where $B \sim B_{\text{crit}}$), is exactly where both the topological and the ultraviolet problem are worst. We did not attempt a magnetised vacuum-seeded run for that reason, and we regard it as the sharpest open question raised by this work: the counterterm we derived is a zero-field tadpole, and its magnetised generalisation must involve the Landau level sum rather than a free-photon sum.

5.7 Cost, and what it implies for three dimensions

All runs reported here are one- or two-dimensional and fit in minutes to hours on 32 CPU cores. The scaling is unforgiving: the resolution constraints $m\Delta t \ll 1$, $\Delta t < \Delta x$ and $qA\Delta x < 2\pi$ fix Δx in units of the Compton wavelength, so a three-dimensional box of physical size L costs $(L/\Delta x)^4$ operations. For the magnetised problem the additional constraint $l_B \gtrsim 5\Delta x$ ties Δx to B_0 , and the flux quantisation (10) ties the smallest representable field to L^{-2} ; the two together mean that the dynamic range in B_0 available at fixed cost is $N_\Phi \in [1, (L/5\Delta x)^2]$. This is a genuine design constraint for any future 3D implementation and, to our knowledge, has not been stated before.

5.8 Questions this work raises but does not answer

Three stand out. (i) Is there a variational modification that makes the classical vacuum stationary, rather than merely cutoff-independent? A gauge-covariant Keldysh thermostat — noise plus dissipation added to the closed-time-path action so that the vacuum spectral weight is the stationary distribution — would do it in principle, at the cost of relaxing exact energy conservation to conservation in the ensemble mean and exact charge conservation to conservation in the mean; we did not implement it. (ii) Does the twisted construction extend to the non-Abelian case relevant for quark–gluon plasmas, where ’t Hooft’s original electric and magnetic fluxes are $\mathbb{Z}_N$ -valued? (iii) Can the dressed-mass result be turned into a diagnostic — the pair gap at $2m\sqrt{1+a^2}$ is a direct, in-situ measurement of the local intensity parameter a , which is notoriously hard to determine in laser–plasma experiments.

6 Conclusion

We have mapped, and in two places repaired, the regime of validity of variational real-time lattice scalar QED as a tool for relativistic-quantum plasma physics. Re-deriving the discretisation from the action rather than from the published update formulas, we confirmed that its advertised invariants are exact — local charge conservation to 1.5×10^{-15} , the Gauss law to 1.5×10^{-14} , the Bianchi identities to 3.8×10^{-16} and gauge invariance to 2.8×10^{-14} — and we reproduced both published demonstrations, recovering the three linear-wave branches to 0.92% and the wakefield-to-Schwinger transition with the published energy-conservation figures.

Within that verified implementation, two new regimes were shown to be quantitatively accessible. A plasma dressed by a relativistically strong circularly polarised wave follows the Volkov mass $m_* = m\sqrt{1+a^2}$ up to $a = 8$ to 0.20%, and with it a simultaneous downward shift of the electromagnetic cutoff (to 0.98%) and upward shift of the pair-creation gap (to 0.73%). Counter-streaming relativistic boson beams reproduce the growth rate of the quantum two-stream instability to 1.08%, including 3 wavenumbers that are unstable in the scalar-QED theory and exactly stable in the relativistic cold-fluid theory: quantum recoil widens the unstable band by 75% and the lattice resolves it. Both regimes exercise the matter–gauge coupling far outside the linear-response corner in which the scheme was originally demonstrated.

Two regimes were shown to be structurally inaccessible, and both were repaired at the level of the discrete action. A uniform magnetic field cannot be carried by any single-valued periodic gauge potential — the plaquette sum of B_z telescopes to exactly zero — so the entire relativistic Landau ladder is absent, with level errors of 5.1% to 10.9% that are independent of the lattice spacing. Restoring ’t Hooft twisted boundary conditions recovers the ladder to 0.18% at every resolution and in two different gauges, preserves exact charge conservation and the Bianchi identities, and relaxes only strict periodicity and the continuous tunability of B_0 , which becomes quantised as $qB_0L_xL_y = 2\pi N$. Separately, seeding every lattice mode with half a quantum — the prescription the classical-statistical approximation requires in order to describe pair creation from the vacuum — gives the charged boson a Rayleigh–Jeans self-energy that diverges logarithmically with the cutoff, so the physical mass drifts by 10.62% and the theory has no continuum limit; a band-limited fluctuation sector plus a gauge-invariant lattice mass counterterm restores it to 0.15% while preserving every exact invariant.

Limitations. Six deserve explicit statement. *First*, the vacuum repair is partial. It makes observables cutoff-independent and restores the Nikishov plateau to 3%–11% at $qE \geq 2m^2$, but the band-limited classical vacuum is still a real classical medium ($\langle |\phi|^2 \rangle = 0.45$, $\omega_p^{\text{vac}} = 0.28m$ at $\Lambda = 8m$), and at the canonical Schwinger point $qE = m^2$ the pair signal is comparable to that background: the measured yield there depends on which modes are averaged, moving over 0.29, 0.59, 0.92, 1.16 as the mode window is widened. We document in Sec. 4.5 and in the attempts log why no counterterm can remove this and what would be required instead. *Second*, all results are 1D or 2D. Three-dimensional runs at these resolutions were outside our compute budget, and while the topological obstruction and the tadpole both have straightforward 3D analogues (the second Chern class and a linearly divergent tadpole respectively), neither has been verified there. *Third*, scalar QED has no spin. Landau levels, pair-creation rates and two-stream growth rates all acquire spin-dependent corrections in spinor QED, and nothing here bears on the fermion-doubling problem that a spinor implementation would face.

Fourth, the dressed-plasma prediction covers the transverse and pair branches only; the Langmuir branch is parametrically mixed with the pump and we report it without a theory (Sec. 5). *Fifth*, our ensembles (8 to 200 realisations, depending on the test) are for some runs smaller than the 200 used in [1], and the $\tau_\omega = 1\%$ tolerance is set by that ensemble noise floor rather than by any intrinsic limit of the method; a larger ensemble would tighten every agreement statement in the paper by roughly $\sqrt{R}$. *Sixth*, the dressed-plasma branch measurement is made in a box small enough that the modulational band of the dressed state is not resolved. That is a deliberate and disclosed choice — the dressed state is not stable in a larger box (Sec. 5) — but it means the probe dispersion is verified only for wavenumbers above the box scale $2\pi/L = 0.61\,m$, and we have not computed the modulational growth rate analytically.

Outlook. The organising principle we extract is that the scheme is exact wherever the gauge configuration is topologically trivial and the matter occupation numbers are large, and that both of its failures are violations of exactly one of those two conditions. That is an encouraging conclusion, because both conditions are diagnosable in advance — a holonomy and an occupation number — and because both repairs are modifications of the discrete Lagrangian rather than of the solver. It also identifies the sharpest open target: magnetised vacuum QED, where the two obstructions compound and where the counterterm must be generalised from a free-photon tadpole to a Landau level sum.

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Appendix

This appendix contains supporting derivations only. Every result claimed in this paper is complete in the sense stated in the main text; the one limitation that we were unable to remove — the residual systematic in absolute Schwinger yields after the vacuum repair — is stated in Sec. 4.5 and Sec. 5 rather than deferred here, and every approach we tried is recorded in `proposal/attempt_s_log.md`.

A Cold multi-beam electrostatic dispersion relation

Write $\phi_z = e^{i(p_z x - E_z t)}(\phi_{0z} + \psi_z)$ with $E_z = \sqrt{m_z^2 + p_z^2}$ and perturb the longitudinal gauge field, $A^x = ae^{i(kx - \omega t)} + \text{c.c.}$ Substituting into $\partial_t^2 \phi - (\partial_x + iqA^x)^2 \phi + m^2 \phi = 0$ and keeping first-order terms gives, for each species,

$$\partial_t^2 \psi - 2iE\partial_t \psi - \partial_x^2 \psi - 2ip\partial_x \psi = iq(\partial_x A^x)\phi_0 - 2qpA^x\phi_0, \quad (22)$$

whose $e^{i(kx - \omega t)}$ component and the corresponding component of the conjugate equation give

$$\psi_+ = \frac{qa\phi_0(k + 2p)}{D_+}, \quad \psi_- = -\frac{qa\phi_0(k - 2p)}{D_-}, \quad D_{\pm} = \omega^2 \pm 2E\omega - k^2 \mp 2pk. \quad (23)$$

The variational current (7) has continuum limit $j^x = 2q(\text{Im}[\bar{\phi}\partial_x \phi] + qA^x|\phi|^2)$, whose $e^{i(kx - \omega t)}$ amplitude is $2q[p\phi_0(\psi_+ + \psi_-) + \frac{k}{2}\phi_0(\psi_+ - \psi_-)] + 2q^2a\phi_0^2$. Ampère's law $\partial_t E^x = j^x$ with $E^x = -\partial_t A^x$ gives $\omega^2 a = \sum_z j_z^x$, i.e. Eq. (16). Setting $p = 0$ and one species reduces the result to $\omega^2 = \omega_p^2[1 + k^2(\omega^2 - k^2)/((\omega^2 - k^2)^2 - 4m^2\omega^2)]$, which is algebraically identical to $1 + \chi_p = 0$ with the published cold susceptibility [78, 1] and, after clearing denominators, to Eq. (11). Two numerical checks were run before the relation was used: the single-species roots agree with the published form to 10^{-15} , and the single-beam roots at drift p are the exact Lorentz transform of the rest-frame roots, again to 10^{-15} — as they must be, because $\tilde{\omega}_p^2 = 2q^2|\phi_0|^2$ is built from the Lorentz scalar $|\phi|^2$.

B Linear response about a circularly polarised pump

Let $A^y + iA^z = A_0 e^{i\omega t}$ with A_0 constant and $k = 0$, so that $A_{\perp}^2 = A_0^2$. The discrete Klein–Gordon equation (4) for a homogeneous ϕ then has no hopping contribution and reduces to the free lattice equation with $m^2 \rightarrow m^2 + q^2 A_0^2$, giving $4\sin^2(m_*\Delta t/2)/\Delta t^2 = m^2 + q^2 A_0^2$, i.e. Eq. (13) in the continuum limit; simultaneously the transverse Ampère law (6) reduces to $4\sin^2(\omega\Delta t/2)/\Delta t^2 = 2q^2|\phi_0|^2$. The pair is an exact solution of the *discrete* system. Since the discrete charge density is $\rho = -(2q/\Delta t)\sin(m_*\Delta t)|\phi_0|^2$, fixing the density n fixes $|\phi_0|^2 = n\Delta t/(2\sin(m_*\Delta t)) \rightarrow n/2m_*$ and hence $\tilde{\omega}_p^2 = q^2 n/m_* = \omega_p^2 m/m_*$, which is Eq. (14).

Perturbing $\phi = e^{-im_* t}(\phi_0 + \psi)$ and $A^x = ae^{i(kx - \omega t)}$ repeats the algebra of App. A with the replacement $m \rightarrow m_*$, because the constant $q^2 A_0^2$ is absorbed into m_*^2 exactly; the longitudinal response is therefore Eq. (11) with $(m, \omega_p) \rightarrow (m_*, \tilde{\omega}_p)$. For a transverse probe δA^y at $k \neq 0$ the wave equation is $\partial_t^2 \delta A^y - \partial_x^2 \delta A^y = -2q^2|\phi|^2 \delta A^y - 2q^2 A^y \delta(|\phi|^2)$; dropping the second term gives (D2). That second term is exactly the parametric coupling discussed in Sec. 5: it is first order in the probe and first order in the pump, so it feeds the transverse probe into density perturbations at $\omega \pm \tilde{\omega}_p$. The pair branch at $2m_*$ is spectrally far from those sidebands and is therefore unaffected, whereas the Langmuir branch at $\tilde{\omega}_p$ sits on top of them; this is why we pre-state (D1)–(D3) but not a Langmuir prediction.